



Faculty of Engineering and Technology
Electrical and Computer Engineering Department
DATA BASE SYSTEMS
COMP333
PHASE 1

I-Tech Electronics Store Management System

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Date: 2026/4/4

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Project Scope and Description

1. System Overview

This project aims to develop a database application for **I-Tech**, a local electronics store that sells a variety of products such as smartphones, laptops, and accessories. The system is designed to support and organize the store's daily operations in an efficient and structured way.

2. Core System Functions

The application manages product information, including product details, categories, and brands. It also handles supplier data, customer records, sales transactions, and purchase transactions. The system allows the store to track stock quantities and monitor product availability.

3. Storage and Organization

The store includes different storage locations, such as a front display area (showroom), storage room, and other areas, which are organized into multiple storage sections. These sections represent different physical areas used to store and arrange products. Each product is assigned to a specific section to ensure better organization and easier access within the store.

4. Warranty Management

The system supports product warranties. Some products, especially electronic devices such as smartphones and laptops, may include a warranty, while other products (such as certain accessories) may not. Warranty information is stored and managed separately and linked only to applicable products.

5. Installment Payments

The system may also support installment payments when needed, allowing the store to manage partial payments over time.

6. Technology and Objectives

The application will be implemented using **Java** as the application layer and **MySQL** as the database management system. It aims to reduce errors in tracking product quantities and simplify daily store operations through organized records, structured relationships, and advanced queries.

System Requirements

1. Functional Requirements

The system should provide the following core functionalities:

- Add, update, delete, and view products
- Manage product categories, suppliers, and customers
- Record purchase transactions and sales transactions
- Track inventory quantities
- Generate reports about stock, sales, and purchases

2. Non-Functional Requirements

The system should satisfy the following quality and performance requirements:

- The system should be user-friendly
- The system should be accurate and reliable
- The system should connect properly to MySQL
- The system should support a graphical user interface using JavaFX
- The system should handle daily store operations efficiently

Information about the Client

1. Client Overview

The client is a local electronics store called I-Tech, located in one of the towns in Ramallah. The store specializes in selling a variety of electronic products such as smartphones, laptops, chargers, headphones, keyboards, mice, and other accessories.

2. Contact Information

The store can be contacted through the following channels:

- Phone: (+970-567200596)
- Email: itech@gmail.com
- Facebook Page: <https://www.facebook.com/share/1EbkdvxVcq/?mibextid=wwXIfr>

3. Client Needs

The client requires a system that can efficiently organize product records, manage supplier and customer information, monitor stock levels, record purchase and sales transactions, and generate useful reports about inventory and business operations.

NOTE: The entities, attributes, and relationships presented in this phase are preliminary and are intended to represent the initial conceptual design of the system. They are not fixed and may be modified, added, or refined later according to the project logic, business rules, and implementation requirements

Expected entities

1. Product

Attributes:

- ProductID (PK)
- ProductName
- Brand
- Model
- Price
- QuantityInStock

2. Category

Attributes:

- CategoryID (PK)
- CategoryName

Examples:

- Smartphone
- Laptop
- Accessory

3. Supplier

Attributes:

- SupplierID (PK)
- SupplierName
- Phone
- Email
- Address
- company

4. Customer

Attributes:

- CustomerID (PK)
- CustomerName
- Phone
- Email

5. Storage Section

Attributes:

- SectionID (PK)
- SectionName
- SectionType

6. Purchase

Attributes:

- PurchaseID (PK)
- PurchaseDate
- TotalAmount

7. Sale

Attributes:

- SaleID (PK)
- SaleDate
- TotalAmount

8. Employee

Attributes:

- EmployeeID (PK)
- EmployeeName
- Position
- Salary
- Phone

9. Brand

- BrandID (PK)
- BrandName
- Country
- Website

10. Warranty (Weak Entity)

Attributes:

- ProductID (Owner Key)
- WarrantyNO(Partial Key)
- WarrantyType
- WarrantyPeriod
- StartDate

11. SaleInstallment (Weak Entity)

Attributes:

- SaleID (Owner Key)
- InstallmentNo (Partial Key)
- Amount
- DueDate
- PaymentStatus
- PaymentDate

Expected Relationships

1. Category — Product

- Each category can include many products.
 - Each product belongs to exactly one category.
- Type: One-to-Many

2. Supplier — Product

- Each supplier provides many products.
 - Each product is supplied by one supplier.
- Type: One-to-Many

3. Customer — Sale

- A customer can make many sales transactions.
 - Each sale is made by one customer.
- Type: One-to-Many

4. Sale — Product

- A sale can include multiple products.
 - A product can appear in multiple sales.
- Type: Many-to-Many

5. Purchase — Product

- A purchase can include multiple products.
 - A product can appear in multiple purchases.
- Type: Many-to-Many

6. Supplier — Purchase

- A supplier can be associated with many purchase transactions.
 - Each purchase is linked to one supplier.
- Type: One-to-Many

7. StorageSection — Product

- Each storage section can contain many products.
 - Each product is stored in one storage section.
- Type: One-to-Many

8. Employee — Employee (Recursive)

- An employee can supervise multiple employees.
 - Each employee is supervised by one manager (or may not have a supervisor).
- Type: One-to-Many (Recursive Relationship)

9. Brand — Product

- Each brand can produce many products.
 - Each product belongs to one brand.
- Type: One-to-Many

10. Product — Warranty

- Each product can have one warranty.
- Each warranty belongs to one product.
Type: One-to-One
- **Participation:** Partial on the Product side and total on the Warranty side.

11. Sale — SaleInstallment

- Each sale can have multiple installments.
- Each installment belongs to exactly one sale.
Type: One-to-Many

Database Design Analysis

1.Total Participation

- **Product** has total participation in:
 - Category — Product (each product must belong to a category).
- **Sale** has total participation in:
 - Customer — Sale (each sale must be linked to a customer).
- **Purchase** has total participation in:
 - Supplier — Purchase (each purchase must be linked to a supplier).
- **Product** may have total participation in:
 - Storage Section — Product (each product must be stored in one section).

2. Recursive Relationship

- The system includes a recursive relationship in:
 - **Employee — Employee**
- An employee can supervise multiple employees.
- Each employee may or may not have a supervisor.
- Type: One-to-Many (recursive)

Participation: Partial

3.Non-Binary Relationships

The current system does not require ternary relationships because all requirements can be represented using binary relationships.

4. Class Hierarchy

Although class hierarchies were considered, the system uses a Category entity to classify products, which provides a simpler and more suitable design for this project.

Sample Queries

1. Retrieve product names, categories, brands, and prices for all products, sorted by category and product name.
2. Retrieve product names, brands, prices, and stock quantities of all products supplied by a specific supplier, sorted by product name.
3. Retrieve total stock quantity for each category, sorted by total quantity in descending order.
4. Retrieve product names, categories, brands, and stock quantities of all products with low stock quantity, sorted by stock quantity.
5. Retrieve sale IDs, customer names, sale dates, and total amounts for all sales made on a specific date, sorted by total amount.
6. Retrieve product names, categories, brands, and stock quantities of all products stored in a specific storage section, sorted by product name.
7. Retrieve product names and total sold quantities of the most sold products, sorted by sold quantity in descending order.
8. Retrieve customer names and the total value of their sales transactions.
9. Retrieve product names, categories, brands, and prices of products that were never sold, sorted by product name.
10. Retrieve the average price of products in each category, sorted by average price in descending order.
11. Retrieve total sales amount per day.
12. Retrieve purchase ID, purchase date, supplier name, product names, quantities, and total amount for a specific purchase transaction.
13. Retrieve product names, categories, models, prices, and stock quantities of all products of a specific brand, sorted by category and price.
14. Retrieve the top 5 brands with the highest number of products.
15. Retrieve sale IDs, sale dates, and total amounts for all sales made by a specific customer, sorted by sale date.
16. Retrieve supplier names, company names, phone numbers, and email addresses of all suppliers, sorted by company name.
17. Retrieve product names, categories, brands, and prices of all products with price less than a specific value, sorted by price.
18. Retrieve employee names, positions, salaries, and phone numbers of all employees supervised by a specific manager, sorted by employee name.
19. Retrieve the average salary of all employees.
20. Retrieve employee names, positions, and salaries of employees who earn above the average salary, sorted by salary in descending order.
21. Retrieve the most expensive product in each category, including product name, category, and price.

